9

Lattice and Carrier Temperature Equations

This chapter describes the equations for lattice and carrier temperatures.

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Introduction to Temperature Equations

Sentaurus Device can compute up to three different temperatures: lattice temperature, electron temperature, and hole temperature. Lattice temperature describes self-heating of devices. The electron and hole temperatures are required to model nonequilibrium effects in deep-submicron devices (in particular, velocity overshoot and impact ionization).

The following options are available:

- The lattice temperature can be computed from the total dissipated heat, assuming the temperature is constant throughout the device.
- The lattice temperature can be computed nonuniformly using the simple model, or the thermodynamic model, or the hydrodynamic model.
- One or both carrier temperatures can be computed using the hydrodynamic model.

Irrespective of the model used, the lattice temperature is stored in the dataset `Temperature`, and electron and hole temperatures are stored as `eTemperature` and `hTemperature`, respectively.

To solve the lattice temperature, you must specify a thermal boundary condition (see Thermal Boundary Conditions on page 316). To solve the carrier temperatures, no thermal boundary conditions are required.

By default (and as an initial guess), the lattice temperature is set to 300 K throughout the device or to the value of `Temperature` set in the `Physics` section. If the device has one or more defined thermodes, an average temperature is calculated from the temperatures at the thermodes and is set throughout the device. As an initial guess, electron and hole temperatures are set to the lattice temperature. Carrier temperatures that are not solved are assumed to equal the lattice temperature throughout the simulation.

Uniform Self-Heating

A simple self-heating model is available to account for self-heating effects without solving the lattice heat flow equation. The self-heating effect can be captured with a uniform temperature, which is bias dependent or current dependent. The global temperature is computed from a global heat balance equation where the dissipated power P_{diss} is equal to the sum of the boundary heat fluxes (on thermodes with finite thermal resistance):

Equation 70

$$P_{\mathrm{diss}} = \sum_i \frac{T - T^{(i)}_{\mathrm{thermode}}}{R^{(i)}_{\mathrm{th}}} + \sum_k \frac{T - T^{(k)}_{\mathrm{circ}}}{R^{(k)}_{\mathrm{th}}}$$

where:

- T is the device global temperature.
- $T^{(i)}_{\mathrm{thermode}}$ and $R^{(i)}_{\mathrm{th}}$ are the temperature and thermal resistivity of thermode i not connected in a thermal circuit.
- $T^{(k)}_{\mathrm{circ}}$ and $R^{(k)}_{\mathrm{th}}$ are the temperature and thermal resistivity of thermode k connected in a thermal circuit.

The second sum in Equation 70 becomes zero when the device is not connected to any thermal circuit.

For a thermode k connected in a thermal circuit, an additional thermal circuit equation is solved for the temperature at the respective thermode $T^{(k)}_{\mathrm{circ}}$. The thermal circuit equation solved is derived from the condition that the heat flux through the connected thermode is equal to the flux flowing through the thermal circuit element to which the thermode is connected.

In the case of thermode k connected to a thermal resistor, the thermal circuit equation becomes:

Equation 71

$$\left(T - T_{\text{circ}}^{(k)}\right) / R_{\text{th}}^{(k)} = \nabla T^{\text{resistor}} / R_{\text{th}}$$

where:

- The left-hand side represents the thermal flux at thermode k.
- The right-hand side represents the heat flux flowing through the thermal resistor with thermal resistivity R_{th}.
- $\nabla T^{\text{resistor}} = T_{\text{resistor}} - T_{\text{circ}}^{(k)}$ is the temperature drop across the thermal resistor.

For a thermal capacitor, the thermal equation becomes

$$\left(T - T_{\text{circ}}^{(k)}\right) / R_{\text{th}}^{(k)} = C_{\text{th}} \frac{d}{dt} \nabla T^{\text{capacitor}}$$

where $\nabla T^{\text{capacitor}} = T_{\text{capacitor}} - T_{\text{circ}}^{(k)}$ is the temperature drop across the thermal capacitor in a transient simulation.

If a thermode does not specify a thermal resistance or a thermal conductance, then by default, a Dirichlet boundary condition of a fixed thermode temperature is assumed. In this case, the thermode is not accounted for in the sum on the right-hand side of Equation 70 because the respective heat flux is zero. When none of the thermodes specifies any thermal resistance or conductance, Equation 70 cannot be solved.

The dissipated power P_{diss} can be calculated as either the sum of all terminal currents multiplied by their respective terminal voltages ($\sum IV$), the sum of *IV*s from the user-specified node, or the total Joule heat $\int_V \left(\vec{J} \cdot \vec{F}\right) dv$.

The global temperature T_i for the point t_i (in transient or quasistationary simulations) is computed based on the estimation of T_i at the previous point t_{i-1} and the solution temperature T_{i-1} at the same point t_{i-1}.

Note:

For uniform self-heating, the temperature is obtained by postprocessing, rather than computed self-consistently with all other solution variables. Therefore, the simulation results might depend on the simulation time step used in quasistationary or transient simulations. In addition, uniform self-heating must not be used for AC, noise, or harmonic balance simulations.

Feedback

Using Uniform Self-Heating

The uniform self-heating equation is activated in the global `Physics` section using the `PostTemperature` keyword:

```
Physics {
   PostTemperature
   ...
}
```

The heat fluxes at thermodes are defined in the `Thermode` sections by specifying `Temperature` and `SurfaceResistance` or `SurfaceConductance`:

```
Thermode {
   {Name= "top" Temperature=300 SurfaceConductance=0.1}
   ...
   {Name= "bottom" Temperature=300 SurfaceResistance=0.01}
}
```

The way P_{diss} is computed can be chosen by options to `PostTemperature` in the command file:

- As $\sum IV$ over all electrodes:

```
Physics {
   PostTemperature(IV_diss)
   ...
}
```

- As $\sum IV$ at user-selected electrodes:

```
Physics {
   PostTemperature(IV_diss("contact1" "contact2"))
   ...
}
```

- As the integral of Joule heat over the device volume:

```
Physics {
   PostTemperature
   ...
}
```

Uniform self-heating can be used during a quasistationary or transient simulation. As the feature replaces the lattice heat flow equation (Equation 72, Equation 76, and Equation 82), it cannot be used with the lattice temperature equation activated in the `Coupled` section.

Simple Model for Lattice Temperature

Sentaurus Device can compute a spatially dependent lattice temperature even when neither the thermodynamic nor the hydrodynamic model is activated. The equation for the temperature is:

Equation 72

$$\frac{\partial W_{\mathrm{L}}}{\partial t} - \nabla \cdot \vec{S}_{\mathrm{L}} = \frac{1}{q}\vec{J}_n \cdot \nabla E_{\mathrm{C}} + + \frac{1}{q}\vec{J}_p \cdot \nabla E_{\mathrm{V}} + \left.\frac{dW_L}{dt}\right|_{\mathrm{coll}}$$

with the energy density and the energy flux as

Equation 73

$$W_L = c_L T \qquad \text{and} \qquad \vec{S}_L = \kappa_L \nabla T$$

the Joule heat terms, $\frac{1}{q}\vec{J}_n \cdot \nabla E_C$ and $\frac{1}{q}\vec{J}_p \cdot \nabla E_V$, and finally the collision term that describes the energy gain/loss due to the activated generation–recombination processes:

Equation 74

$$\left.\frac{dW_L}{dt}\right|_{coll} = H_L = E_{g,eff} R_{total} + (\hbar\omega - E_{g,eff}) G^{opt}$$

where:

- $E_{g,eff}$ is the effective band gap.
- R_{total} describes the total recombination rate.
- $\hbar\omega$ is the photon energy.
- G^{opt} is the optical generation rate.

Usually, the influence of H_L is small, so it is not activated by default. To take this energy source into account, the keyword `RecGenHeat` keyword must be specified in the `Physics` section.

In metals, Equation 72 degenerates into:

Equation 75

$$\frac{\partial W_{\mathrm{L}}}{\partial t} - \nabla \cdot \vec{S}_L = -\nabla \Phi_M \cdot \vec{J}_M$$

where:

- Φ_M is the Fermi potential in the metal.
- $\boldsymbol{J}_M$ is the metal current density as defined in Equation 147.

Using the Simple Model for Lattice Temperature

Because the thermodynamic model (see Thermodynamic Model for Lattice Temperature) is used by default, you must specify `-Thermodynamic` in the `Physics` section together with the `Temperature` keyword in the `Solve` section to use the simple model for lattice temperature.

To account for Peltier heat at a contact, or a metal–semiconductor interface, or a conductive insulator–semiconductor interface, switch on the Peltier heating model explicitly. See Heating at Contacts, Metal–Semiconductor and Conductive Insulator–Semiconductor Interfaces on page 1150.

Thermodynamic Model for Lattice Temperature

With the thermodynamic model, the lattice temperature T is computed from:

Equation 76

$$\begin{aligned}\frac{\partial}{\partial t}(c_{\mathrm{L}}T) - \nabla\cdot(\kappa_{\mathrm{L}}\nabla T) \quad = \quad & -\nabla\cdot\left[(P_nT+\Phi_n)\vec{J}_n + (P_pT+\Phi_p)\vec{J}_p\right] \\ & -\tfrac{1}{q}\left(E_{\mathrm{C}}+\tfrac{3}{2}kT\right)\left(\nabla\cdot\vec{J}_n - qR_{\mathrm{net},n}\right) \\ & -\tfrac{1}{q}\left(-E_{\mathrm{V}}+\tfrac{3}{2}kT\right)\left(-\nabla\cdot\vec{J}_p - qR_{\mathrm{net},p}\right) + \hbar\,\omega\,G^{\mathrm{opt}}\end{aligned}$$

where:

- κ_{L} is the thermal conductivity (see Thermal Conductivity on page 1131).
- c_{L} is the lattice heat capacity (see Heat Capacity on page 1129).
- E_{C} and E_{V} are the conduction and valence band energies, respectively.
- G^{opt} is the optical generation rate from photons with frequency ω (see Chapter 21, Optical Generation on page 729).
- $R_{\mathrm{net},n}$ and $R_{\mathrm{net},p}$ are the electron and hole net recombination rates, respectively.
- Current densities J_n and J_p are computed as described in Thermodynamic Model for Current Densities on page 269.

In metals, Equation 76 degenerates into:

Equation 77

$$\frac{\partial}{\partial t}(c_{\mathrm{L}}T) - \nabla\cdot(\kappa_{\mathrm{L}}\nabla T) = -\nabla\cdot\left[(PT+\Phi_{\mathrm{M}})\vec{J}_M\right]$$

where:

- P is the metal thermoelectric power.
- Φ_{M} is the Fermi potential in the metal.
- $\vec{J}_M$ is the metal current density as defined in Equation 147.

Total Heat and Its Contributions

The RHS of Equation 76 is the total heat H. In the stationary case, the second term and the third term disappear, that is, the total heat is then given by:

Equation 78

$$H = -\nabla\cdot\left[(P_nT+\Phi_n)\vec{J}_n + (P_pT+\Phi_p)\vec{J}_p\right] + \hbar\omega G^{\mathrm{opt}}$$

The electron part can be rewritten as:

Equation 79

$$-\nabla\cdot\left[(P_nT+\Phi_n)\vec{J}_n\right] = \frac{1}{qn\mu_n}\left|\vec{J}_n\right|^2 - qT\nabla P_n\cdot\vec{J}_n + \left[(P_nT+\Phi_n)\vec{J}_n\cdot n_S\right]_S\delta_S - q(P_nT+\Phi_n)R_{\mathrm{net},n}$$

where:

- n_S is the surface normal at region interface S.
- δ_S denotes the surface delta function at region interface S.
- $[\alpha]_S$ denotes the jump of a function α across the region interface S.

The resulting four terms are often denoted as the electron part of the Joule heat, Thomson heat, Peltier heat, and recombination heat, respectively.

Using the Thermodynamic Model

To use the thermodynamic model, you must specify the `Temperature` keyword in the `Coupled` statement of the `Solve` section. You need not specify a model in the `Physics` section because the thermodynamic model is used by default. However, optionally, the `Thermodynamic` keyword can still be specified in the `Physics` section.

The heating mechanisms of nonlocal line tunneling are not accounted for, by default, in the thermodynamic model. Therefore, to also take into account heating by nonlocal line tunneling, you must add the `NLLTunnelingRecGenHeat` option to the `Thermodynamic` keyword as `Thermodynamic(NLLTunnelingRecGenHeat)`.

To account for Peltier heat at a contact, or a metal–semiconductor interface, or a conductive insulator–semiconductor interface, switch on the Peltier heating model explicitly (see Heating at Contacts, Metal–Semiconductor and Conductive Insulator–Semiconductor Interfaces on page 1150).

Plotting Options for the Lattice Temperature Models

To plot the lattice heat flux vector $\kappa_L \nabla T$, specify the `lHeatFlux` keyword in the `Plot` section (see Device Plots).

Sentaurus Device allows the total heat generation rate and the separate components of the total heat to be plotted. The total heat generation rate is the term on the right-hand side of Equation 72 or Equation 76. It is plotted using the `TotalHeat` keyword in the `Plot` section. The total heat and the separate components are calculated only when Sentaurus Device solves the temperature equation.

Table 31 shows the formulas for the individual heating mechanisms and the appropriate keywords to use in the `Plot` section of the command file (see Device Plots). A summary of the individual heat terms and the total heat is also written to the `.log` file.

In addition, Sentaurus Device reports the maximum, minimum, and average device temperatures in both the plot file and log file whenever the self-heating equation is solved:

- Maximum temperature (T_{max}) in the device as a function of the bias condition
- Minimum temperature (T_{min}) in the device as a function of the bias condition
- Average temperature (T_{avg}) in the device as a function of the bias condition

Device temperatures are not monitored and remain the same as the default lattice temperature, that is, room temperature, if the lattice temperature solution is not requested.

You can also use the current plot feature of Sentaurus Device to extract temperature information in different materials and regions. For example:

```
CurrentPlot {
         Temperature(
              Minimum(Material="Silicon")
              Maximum(Material="Silicon")
              Average(Material="Silicon")
              Minimum(Material="Oxide")
              Maximum(Material="Oxide")
              Average(Material="Oxide")
```

```
            )
        }
```

Table 31 Terms and keywords used in Plot section of command file

Heat name	Keyword	Formula Simple Model	Formula TD Model
Electron Joule heat	`eJouleHeat`	$\frac{1}{q}\vec{J}_n \cdot \nabla E_C$	$-\left(P_n \nabla T + \nabla \Phi_{F,n}\right) \cdot \vec{J}_n$
Hole Joule heat	`hJouleHeat`	$\frac{1}{q}\vec{J}_p \cdot \nabla E_V$	$-\left(P_p \nabla T + \nabla \Phi_{F,p}\right) \cdot \vec{J}_p$
Joule heat in metals and conductive insulators	`JouleHeat`	$\sigma \left\| \nabla \Phi_M \right\|^2$	$\sigma \left\| \nabla \Phi_M + P_M \nabla T \right\|^2$
Recombination heat	`1RecGenHeat`	$E_{g,eff} R_{total}$	$qR_{total}\left(\left(TP_p + \Phi_{F,p}\right) - \left(TP_n + \Phi_{F,n}\right)\right)$
Thomson+Peltier heat [1]	`ThomsonHeat`	Not Treated	$-T\nabla P_n \cdot \vec{J}_n - T\nabla P_p \cdot \vec{J}_p$
Peltier heat	`PeltierHeat`	Not Treated	$-T\ \frac{\partial Pn}{\partial n}\nabla n \cdot \vec{J}_n - T\ \frac{\partial P_p}{\partial p}\nabla p \cdot \vec{J}_p$

Hydrodynamic Model for Temperatures

Deep-submicron devices cannot be described properly using the conventional drift-diffusion transport model. In particular, the drift-diffusion approach cannot reproduce velocity overshoot and often overestimates the impact ionization generation rates. In this case, the hydrodynamic (or energy balance) model provides a good compromise between physical accuracy and computation time.

Since the work of Stratton [2] and Bløtekjær [3], there have been many variations of this model. The full formulation includes convective terms [4]. Sentaurus Device implements a simpler formulation without convective terms. In addition to the Poisson equation (Equation 38) and continuity equations (Equation 58), up to three additional equations for the lattice temperature T and the carrier temperatures T_n and T_p can be solved.

The energy balance equations read:

Equation 80

$$\frac{\partial W_n}{\partial t} + \nabla \cdot \vec{S}_n = \vec{J}_n \cdot \nabla E_C / q + \left.\frac{dW_n}{dt}\right|_{coll}$$

Equation 81

$$\frac{\partial W_p}{\partial t} + \nabla \cdot \vec{S}_p = \vec{J}_p \cdot \nabla E_V / q + \left.\frac{dW_p}{dt}\right|_{coll}$$

Equation 82

$$\frac{\partial W_{\mathrm{L}}}{\partial t} + \nabla \cdot \vec{S}_{\mathrm{L}} = \left.\frac{dW_{\mathrm{L}}}{dt}\right|_{\mathrm{coll}}$$

where the energy fluxes are:

Equation 83

$$\vec{S}_n = -\frac{5r_n \lambda_n}{2}\left(\frac{kT_n}{q}\vec{J}_n + f_n^{\mathrm{hf}}\,\hat{\kappa}_n \nabla T_n\right)$$

Equation 84

$$\vec{S}_p = -\frac{5r_p \lambda_p}{2}\left(\frac{-kT_p}{q}\vec{J}_p + f_p^{\mathrm{hf}}\,\hat{\kappa}_p \nabla T_p\right)$$

Equation 85

$$\vec{S}_{\mathrm{L}} = -\kappa_{\mathrm{L}:}\nabla T_{\mathrm{L}}$$

Equation 86

$$\hat{\kappa}_n = \frac{k^2}{q} n \mu_n T_n$$

Equation 87

$$\hat{\kappa}_p = \frac{k^2}{q} p \mu_p T_p$$

and λ_n and λ_p are defined as for Equation 65 and Equation 66.

The parameters r_n, r_p, f_n^{hf}, and f_p^{hf} are accessible in the parameter file of Sentaurus Device. Different values of these parameters can significantly influence the physical results, such as velocity distribution and possible spurious velocity peaks. By changing these parameters, Sentaurus Device can be tuned to a very wide set of hydrodynamic or energy balance models as described in the literature. The default parameter values of Sentaurus Device are:

Equation 88

$$r_n = r_p = 0.6$$

Equation 89

$$f_n^{\mathrm{hf}} = f_p^{\mathrm{hf}} = 1$$

By changing the constants f_n^{hf} and r_n, you can change the convective contribution and the diffusive contributions independently. With the default set of transport parameters of Sentaurus Device, the prefactor of the diffusive term has the form:

Equation 90

$$\kappa_n = \frac{3}{2} \cdot \frac{k^2 \lambda_n}{q} n \mu_n T_n$$

The collision terms are expressed by this set of equations:

Equation 91

$$\left.\frac{\partial W_n}{\partial t}\right|_{\mathrm{coll}} = -H_n - \xi_n \frac{W_n - W_{n0}}{\tau_{\mathrm{e}n}}$$

Equation 92

$$\left.\frac{\partial W_p}{\partial t}\right|_{\mathrm{coll}} = -H_p - \xi_p \frac{W_p - W_{p0}}{\tau_{\mathrm{e}p}}$$

Equation 93

$$\left.\frac{\partial W_{\mathrm{L}}}{\partial t}\right|_{\mathrm{coll}} = H_{\mathrm{L}} + \xi_n \frac{W_n - W_{n0}}{\tau_{\mathrm{e}n}} + \xi_p \frac{W_p - W_{p0}}{\tau_{\mathrm{e}p}}$$

Here, H_n, H_p, and H_{L} are the energy gain/loss terms due to generation–recombination processes. The expressions used for these terms are based on approximations [5] and have the following form for the major generation–recombination phenomena:

Equation 94

$$H_n = 1.5kT_n\left(R_{\mathrm{net}}^{\mathrm{SRH}} + R_{\mathrm{net}}^{\mathrm{rad}} + R_{n,\mathrm{net}}^{\mathrm{trap}}\right) - E_{\mathrm{g,eff}}\left(R_n^{\mathrm{A}} - G_n^{\mathrm{ii}}\right) - \alpha\left(\hbar\omega - E_{\mathrm{g,eff}}\right)G^{\mathrm{opt}}$$

Equation 95

$$H_p = 1.5kT_p\left(R_{\mathrm{net}}^{\mathrm{SRH}} + R_{\mathrm{net}}^{\mathrm{rad}} + R_{p,\mathrm{net}}^{\mathrm{trap}}\right) - E_{\mathrm{g,eff}}\left(R_p^{\mathrm{A}} - G_p^{\mathrm{ii}}\right) - (1-\alpha)\left(\hbar\omega - E_{\mathrm{g,eff}}\right)G^{\mathrm{opt}}$$

Equation 96

$$H_{\mathrm{L}} = \left[R_{\mathrm{net}}^{\mathrm{SRH}} + 0.5\left(R_{n,\mathrm{net}}^{\mathrm{trap}} + R_{p,\mathrm{net}}^{\mathrm{trap}}\right)\right]\left(E_{\mathrm{g,eff}} + 1.5kT_n + 1.5kT_p\right)$$

where:

- $R_{\mathrm{net}}^{\mathrm{SRH}}$ is the Shockley–Read–Hall (SRH) recombination rate (see Shockley–Read–Hall Recombination on page 526).
- $R_{\mathrm{net}}^{\mathrm{rad}}$ is the radiative recombination rate (see Radiative Recombination Model on page 543).

- R_n^{A} and R_p^{A} are Auger recombination rates related to electrons and holes (see Auger Recombination Model on page 544).
- $\hbar\omega$ is the photon energy (see Hydrodynamic Model Parameters on page 286).
- α is a dimensionless parameter that describes how the surplus energy of the photon splits between the bands (see Hydrodynamic Model Parameters on page 286).
- G_n^{ii} and G_p^{ii} are impact ionization rates (see Avalanche Generation on page 549).
- $R_{n,\mathrm{net}}^{\mathrm{trap}}$ and $R_{p,\mathrm{net}}^{\mathrm{trap}}$ are the recombination rates through trap levels (see Chapter 17, Traps and Fixed Charges on page 608).
- G^{opt} is the optical generation rate (see Chapter 21, Optical Generation on page 729).

Surface recombination is taken into account in a way similar to bulk SRH recombination. Usually, the influence of H_n, H_p, and H_{L} is small, so they are not activated by default. To take these energy sources into account, the keyword `RecGenHeat` must be specified in the `Physics` section.

The energy densities W_n, W_p, and W_{L} are given by:

Equation 97

$$W_n = nw_n = n\left(\frac{3kT_n}{2}\right)$$

Equation 98

$$W_p = pw_p = p\left(\frac{3kT_p}{2}\right)$$

Equation 99

$$W_{\mathrm{L}} = c_{\mathrm{L}}T$$

The corresponding equilibrium energy densities are:

Equation 100

$$W_{n0} = nw_0 = n\frac{3kT}{2}$$

Equation 101

$$W_{p0} = pw_0 = p\frac{3kT}{2}$$

The parameters ξ_n and ξ_p in Equation 91 to Equation 93 improve numeric stability. They speed up relaxation for small densities and they approach 1 for large densities:

Equation 102

$$\xi_n = 1 + \frac{n_{\min}}{n}\left(\frac{n_0}{n_{\min}}\right)^{\max[0,(T-T_n)/100\mathrm{K}]}$$

and likewise for ξ_p. Here, $n_{\min}$ and n_0 are adjustable small density parameters.

Hydrodynamic Model Parameters

The default set of transport coefficients (Equation 88 and Equation 89) can be changed in the parameter file. The coefficients r and f^{hf} are specified in the `EnergyFlux` and `HeatFlux` parameter sets, respectively. Energy relaxation times $\tau_{\mathrm{e}n}$ and $\tau_{\mathrm{e}p}$ can be modified in the `EnergyRelaxationTime` parameter set.

α in Equation 94 and Equation 95 divides the contribution of the optical generation rate into the energy gain or loss terms H_n and H_p. `OptGenOffset` specifies α, which can take values between 0 and 1 (default is 0.5). The angular frequency ω in Equation 94 and Equation 95 corresponds to the wavelength specified to compute the optical generation rate. This wavelength is defined by `OptGenWavelength` if the optical generation is loaded from a file (see Optical AC Analysis on page 857).

Both `OptGenOffset` and `OptGenWavelength` are options to `RecGenHeat` (see Table 246 on page 1765).

The parameters $n_{\min}$ and n_0 in Equation 102 are set with `RelTermMinDensity` and `RelTermMinZeroDensity`, respectively, in the global `Math` section. The default values for $n_{\min}$ and n_0 are 10^3 cm^{-3} and 2×10^8 cm^{-3}, respectively.

Using the Hydrodynamic Model

To activate the hydrodynamic model, you must specify the keyword `Hydrodynamic` in the `Physics` section. If only one carrier temperature equation is to be solved, `Hydrodynamic` must be specified with the appropriate parameter, either `Hydrodynamic(eTemperature)` or `Hydrodynamic(hTemperature)`. If the hydrodynamic model is not activated for a particular carrier type, Sentaurus Device merges the temperature for this carrier with the lattice temperature. That is, these temperatures are equal, and their heating terms (see the right-hand sides of Equation 80, Equation 81, and Equation 82) are added.

In addition, you must specify the `eTemperature`, `hTemperature`, and `Temperature` keywords inside the `Coupled` statement of the `Solve` section (see Coupled Statement on page 214) to actually solve a carrier temperature equation or the lattice temperature equation. Temperatures remain fixed during `Solve` statements in which their equation is not solved.

By default, the energy conservation equations of Sentaurus Device do not include generation–recombination heat sources. To activate them, the keyword `RecGenHeat` must be specified in the `Physics` section.

To plot the electron, hole, and lattice heat flux vectors $\vec{S}_n$, $\vec{S}_p$, $\vec{S}_\mathrm{L}$ (see Equation 83, Equation 84, and Equation 85), specify the corresponding keywords `eHeatFlux`, `hHeatFlux`, and `lHeatFlux` in the `Plot` section (see Device Plots on page 195).

To plot $\vec{J}_n \cdot \nabla E_\mathrm{C} / q$ and $\vec{J}_p \cdot \nabla E_\mathrm{V} / q$ (see Equation 80 and Equation 81), specify the corresponding keywords `eJEheat` and `hJEHeat`.

To plot $dW_n / dt|_\mathrm{coll}$, $dW_p / dt|_\mathrm{coll}$, and $dW_\mathrm{L} / dt|_\mathrm{coll}$ (see Equation 80, Equation 81, and Equation 82), specify the corresponding keywords `eColHeat`, `hColHeat`, and `lColHeat`.

To plot H_n, H_p, and H_L (see Equation 94, Equation 95, and Equation 96), specify the corresponding keywords `eRecGenHeat`, `hRecGenHeat`, and `lRecGenHeat`.

Numeric Parameters for Temperature Equations

This section presents the numeric parameters for temperature equations.

Validity Ranges for Lattice and Carrier Temperatures

Lower and upper limits for the lattice temperature and the carrier temperatures exist.

The allowed temperature ranges are specified (in K) by `lT_Range=(<float> <float>)` (defaults 50 K and 5000 K) and `cT_Range=(<float> <float>)` (defaults 10 K and 80 000 K). These ranges apply both to the temperatures during the Newton iterations and to the final results.

Scaling of Lattice Heat Generation

For the lattice heat equation, you can deactivate the heat term by using:

```
Physics ( Region="Bulk" ) { ... HeatPreFactor = 0. }
```

This reduces or even eliminates the strong coupling, given by the Joule heat, of the lattice heat equation with the carrier continuity equations, and leads in general to smooth temperature profiles. This simplified coupled system of carrier transport and lattice temperature might converge easier than the fully coupled system including the heat term. Instead of ramping the fully coupled system to a required bias condition, it might be easier to ramp first the simplified coupled system to the required bias condition, and ramp subsequently the `HeatPreFactor` from zero to one.

Feedback

Note:

Due to the strong nonlinearity of semiconductor equations, which allows multiple solutions in general, and especially for examples with strong generation–recombination rates, the `HeatPreFactor` ramping might lead to a different but still valid solution.

In the command file, this second ramp could be:

```
Quasistationary (
   Goal { Region="Bulk" ModelParameter="Physics/HeatPreFactor" Value=1 }
) { Coupled { Poisson Electron Hole Temperature } }
```

Note:

The `Goal` statement requires the specification of a `Region`, and a `Physics` section for the corresponding region must be specified in the command file.

References

[1] K. Kells, *General Electrothermal Semiconductor Device Simulation*, Series in Microelectronics, vol. 37, Konstanz, Germany: Hartung-Gorre, 1994.

[2] R. Stratton, “Diffusion of Hot and Cold Electrons in Semiconductor Barriers,” *Physical Review*, vol. 126, no. 6, pp. 2002–2014, 1962.

[3] K. Bløtekjær, “Transport Equations for Electrons in Two-Valley Semiconductors,” *IEEE Transactions on Electron Devices*, vol. ED-17, no. 1, pp. 38–47, 1970.

[4] A. Benvenuti *et al.*, “Evaluation of the Influence of Convective Energy in HBTs Using a Fully Hydrodynamic Model,” in *IEDM Technical Digest*, Washington, DC, USA, pp. 499–502, December 1991.

[5] D. Chen *et al.*, “Dual Energy Transport Model with Coupled Lattice and Carrier Temperatures,” in *Simulation of Semiconductor Devices and Processes (SISDEP)*, vol. 5, Vienna, Austria, pp. 157–160, September 1993.